

OBs

Video:

<https://coreultrasound.com/ob-gyn-scanning-technique-5ms/>
<https://coreultrasound.com/intrauterine-pregnancy-assessment-5ms/>
<https://coreultrasound.com/fhr/>

Indications/Goals of Exam:

- Evaluation of pain and bleeding in the first trimester of pregnancy
- *Rule in* an intrauterine pregnancy, remember you cannot absolutely rule out an ectopic pregnancy.
- Identify other relevant pathology

Epidemiology:

- 40% of pregnancies have 1st trimester bleeding with 40-50% of these progressing to miscarriage
- Ectopic rate is about 1/80 in all pregnancies and about 10% of those presenting with 1st trimester bleeding.
- Heterotopic rate is 1/30000, but increasing to 1/100 with IVF/fertility treatments

Pregnancy Failure:

- 1) No gestational sac at BhCG of > 3000 transabdominally (>1500 transvaginal)
- 2) No yolk sac with gestational sac > 15mm
- 3) No fetal pole with gestational sac > 25mm
- 4) CRL > 5-7mm with no FHR
- 5) No FHR after 10-12 weeks gestational age

3-2-1 rule for IUP

- **Pregnancy Criteria (3) - need all to confirm a pregnancy:**
 - o **1. Decidual reaction**
 - Occurs approximately 14th day post-fertilization a decidual reaction occurs and this leads to the formation of a strongly echogenic (white) lining
 - o **2. Gestational sac**
 - Look for an anechoic (black) area contained in the decidual reaction
 - o **3. Yolk sac OR fetal pole with visible fetal heart**
 - **YOLK SAC:**
 - Thick echogenic layer found within the gestational sac
 - o The “double ring sign” - the yolk sac within gestational sac, is the earliest sign of a definitive IUP
 - o Becomes visible when the gestational sac is 10mm by TV and 20mm by AB.
 - Can be seen by **5.5 weeks by TVUS**
 - Can be seen by **6.5 weeks by ABUS**
 - **FETAL POLE:**
 - o Develops at the same time as the yolk sac, but is realistically picked up by EDE 1 week after yolk sac
 - Can be seen by **6 weeks by TVUS**
 - Can be seen **7-8 weeks by ABUS**
 - **Lives IUP:**
 - Must have a **heart rate of > 100** to be consistent with a good fetal outcome.
 - o Reliably seen **after 6 weeks** by TVUS and a week or **after 8 weeks** by ABUS.
 - Cardiac activity should always be detectable if the **fetal pole exceeds 5mm by TV or 10mm by AB.**
 - **Intrauterine criteria (2) - need both to confirm intrauterine pregnancy:**

- Bladder-uterine juxtaposition. At least one image must clearly show the bladder and uterine tissue in direct contact
- Vaginal uterine continuity. In longitudinal view, the vagina must be shown to transverse directly into uterine tissue
- **Safety Criteria (1) - to confirm safe location within the uterus**
 - Adequate myometrial mantle. The shortest distance between the inner edge of the gestational sac and the outer edge of the uterus must be at least 5mm.

Knobology/Preparation

- Curvilinear probe
- Depth: 15cm
- Requires a full bladder (allows for a good acoustic window to look through to see the uterus)
- System preset: OB
- Gain: mid-range

External Landmark: cephalad to symphysis pubis

Internal Landmark: bladder

AOI:

- Pregnancy: **uterus/endometrial stripe**
 - In longitudinal plane: endometrial stripe = centrally along the long axis of the uterus
 - In the transverse plane, the endometrial stripe, when visible, should be circular and located centrally within the uterus.
- Free fluid: **recto-uterine/vesico-uterine spaces**

Steps:

1. Place the probe in the midline (curved array probe) in a **longitudinal** orientation.
 1. This should be done immediately cephalad to the symphysis pubis
 2. External landmark = symphysis pubis
2. Heel the probe caudally until the bladder moves to the top right of screen
 1. Look for the uterus which is a mostly solid organ, so it should resemble the liver/spleen. Will appear farfield to the bladder
 2. May need to troubleshoot at this step as the uterus may live off center.
 1. Slide probe L and R → looking for where the uterus appears the largest.
3. Center the uterus by adjusting depth, heeling probe, etc.
 1. Heeling the probe cephalad will move the uterus right
 2. Heeling the probe caudad will move the uterus left.
4. Sweep through the uterus → should appear as an elongated pear.
 1. Look for endometrial stripe (higher echogenicity that represents the lining of the uterus) → consider endometrial stripe the area of interest, rather than just using the uterus.
5. Rotate probe 90 degrees and sweep again in **transverse** orientation
 1. Area of interest: uterus (circular shape in this orientation)
 1. Should start the scan from the vaginal stripe in this orientation.

Troubleshooting:

- Make sure it is right next to the pubic symphysis
- Press and hold to displace gas
- Slide the probe and heel back → aka move the probe off center and then heel it to point towards the probe.
 - Want the bladder as an acoustic window.
- Reverse trendelenburg for better visibility of pelvic free fluid
- Turn gain down if washout from acoustic window of bladder
- Give fluids to fill the bladder (IV fluids >>>> catheter), reminder that PO fluids is contraindicated in someone who may have a surgical abdomen...

Troubleshooting to find the yolk sac

- Change orientation of scan (transverse → longitudinal or vice versa)
- Increase probe frequency (higher frequency) → can try linear probe
- Consider zooming in on the gestational sac

Pitfalls (interpretation)

- Mistaken pseudo-gestational sac for IUP
- Failure to ensure bladder/uterine juxtaposition
- Mistaking maternal blood flow in the decidual reaction as fetal cardiac activity

Pitfalls Clinical Integration:

- Assuming IUP, despite 3-2-1 rule not being fulfilled
- Mistaking physiologic fluid for blood
- Assuming free fluid in pelvis is physiologic
- Declaring safe IUP with no yolk sac is visible and a presumed fetal pole does not have an obvious fetal heart